

Full OFC State (14D) $\mathbf{x} = [x, y, \alpha, \beta, \dot{x}, \dot{y}, \dot{\alpha}, \dot{\beta}, F_x, F_y, x_{\text{ref}}, y_{\text{ref}}, \dot{x}_{\text{ref}}, \dot{y}_{\text{ref}}]^\top \in \mathbb{R}^{14}$

$$J = \int_0^T [\mathbf{e}^\top \mathbf{Q} \mathbf{e} + \mathbf{u}^\top \mathbf{R} \mathbf{u}] dt + \mathbf{e}(T)^\top \mathbf{Q}_N \mathbf{e}(T)$$

$\mathbf{e} = \mathbf{x} - \mathbf{x}^*$ is the 14D state error with goal state

$$\ddot{\mathbf{p}}_{\text{ref}}(t) = \frac{1}{M_{\text{ref}}} [\mathbf{F}(t) + \mathbf{K}(\mathbf{p}(t) - \mathbf{p}_{\text{ref}}(t)) + \mathbf{D}(\dot{\mathbf{p}}(t) - \dot{\mathbf{p}}_{\text{ref}}(t))]$$

$\mathbf{p} = [x, y]^\top$, $\mathbf{p}_{\text{ref}} = [x_{\text{ref}}, y_{\text{ref}}]^\top$.

